

Graduate Course Syllabus

ME 664 Nonlinear Oscillations

Course format	Master's and PhD level 14 weeks 3 hours per week 42 contact hours Instructor: Saeed Lotfan
Course scope	Modeling and analysis of nonlinear mechanical oscillations in discrete and continuous systems. Emphasis on physical interpretation, perturbation methods, harmonic balance, resonance, stability, modal interactions, and nonlinear model reduction.
Prerequisites	Ordinary differential equations, linear algebra, Fourier series, and linear single- and multi-degree-of-freedom vibrations; basic numerical integration and programming.
Learning outcomes	1. Derive and nondimensionalize nonlinear equations of motion and interpret phase portraits and energy landscapes. 2. Apply perturbation and averaging methods to free, forced, self-excited, and parametrically excited oscillations. 3. Formulate and solve harmonic-balance equations, assess harmonic convergence, and trace periodic-response branches. 4. Analyze stability, bifurcations, internal resonance, and energy exchange in coupled systems. 5. Construct reduced models of continuous systems and explain nonlinear normal modes. 6. Compare analytical and numerical predictions and state their validity and limitations.
Teaching approach	Compact delivery: Linear-vibration review is assigned before class; derivations and numerical examples are integrated each week. Advanced manifold calculations and broader parameter studies form PhD extensions.
Main textbooks	A. H. Nayfeh and D. T. Mook, Nonlinear Oscillations, Wiley, 1979. M. Krack and J. Gross, Harmonic Balance for Nonlinear Vibration Problems, Springer, 2019.
Advanced reading	C. Touze and A. Vizzaccaro, Nonlinear normal modes as invariant manifolds for model order reduction, 2024. O. Thomas, Understanding, computing and identifying the nonlinear dynamics of elastic and piezoelectric structures thanks to nonlinear modes, 2024. Both chapters appear in Model Order Reduction for Design, Analysis and Control of Nonlinear Vibratory Systems, Springer.
Practice and scope	Use the pendulum, Duffing and van der Pol oscillators, Mathieu-type systems, coupled oscillators, and a nonlinear beam as recurring examples. All 14 meetings contain teaching; any in-class examination or project presentations require a separate time allocation.

Topics covered and schedule

Week	Topics and coverage
1	Modeling and qualitative dynamics Nonlinearity and nondimensionalization; conservative and nonconservative systems; equilibria, stability, phase portraits, energy, and separatrices.
2	Conservative oscillations and perturbation Amplitude-dependent period and backbone curves; regular perturbation and secular terms; Poincare-Lindstedt method for quadratic and cubic nonlinearities.
3	Multiple scales and averaging Slow amplitude and phase evolution; solvability conditions; weakly damped free response; comparison of multiple scales, averaging, and direct integration.
4	Self-excited oscillations and limit cycles Van der Pol oscillator; amplitude selection; stability of averaged equilibria and limit cycles; introduction to Hopf bifurcation.
5	Forced primary resonance Quadratic and cubic nonlinearities; frequency-response and phase curves; hardening and softening; folds, multistability, jumps, and hysteresis.
6	Secondary and combination resonances Selected superharmonic and subharmonic resonances; multiple-frequency forcing and combination resonances; scaling, detuning, and modulation equations.
7	Parametric excitation and Floquet theory Mathieu and Hill equations; instability tongues; parametric resonance and nonlinear saturation; variational equations, monodromy matrix, and Floquet multipliers.
8	Harmonic balance fundamentals Fourier representation and weighted residuals; single- and multi-harmonic approximations; static component; Duffing response; residuals and harmonic-truncation convergence.
9	Computational harmonic balance Alternating frequency-time evaluation; Newton iteration and Jacobians; predictor-corrector and pseudo-arclength continuation; tracing folds; checks against integration and Floquet stability.
10	Coupled nonlinear oscillators Modal coordinates and nonlinear coupling; multiple scales for multi-degree-of-freedom systems; internal-resonance conditions; worked 1:2 or 1:3 resonance and energy exchange.
11	Modal interactions and multi-mode response Autoparametric resonance, saturation, and energy transfer; coupled response stability; multi-degree-of-freedom harmonic balance and continuation.
12	Continuous systems and large-amplitude vibration Geometric nonlinearity in beams; Galerkin discretization and modal truncation; reduced equations with quadratic and cubic coupling; spatial and temporal convergence checks.
13	Nonlinear normal modes and model reduction Periodic-orbit and invariant-manifold viewpoints; backbone curves; normal forms and nonlinear modal mappings; low-order reduction example and validity near internal resonance.
14	Advanced dynamics and synthesis Poincare sections, period doubling, and a brief introduction to chaos; integrate perturbation, harmonic balance, continuation, and reduced modeling in a structural-vibration case study.